```
% ========================================================================
% Multi-domain Supercapacitor PDE Model with 2D Voltage & Energy Maps
% ========================================================================
clear; clc; close all;

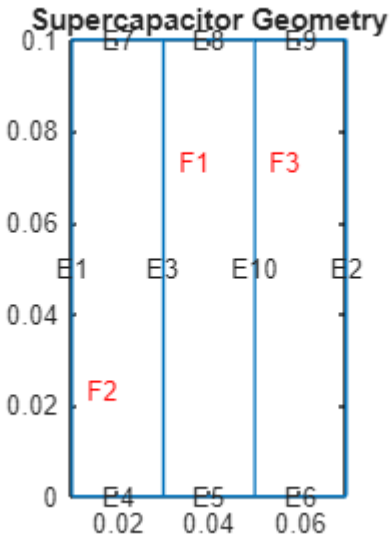
%% 1. PARAMETERS
h_cap = 0.1; l_a = 0.02; l_e = 0.02; l_c = 0.02; r_in = 0.01;
sigma_s = 1e-3; sigma_l = 1e-1;
eps_s = 0.75; eps_l = 0.6;
C_dl = 0.13; A_v = 1.2e6;
I_cell = 0.01;

%% 2. PDE MODEL (scalar PDE)
model = createpde(1);

%% 3. GEOMETRY
r1 = [3;4;r_in;r_in+l_a;r_in+l_a;r_in;0;0;h_cap;h_cap];
r2 = [3;4;r_in+l_a;r_in+l_a+l_e;r_in+l_a+l_e;r_in+l_a;0;0;h_cap;h_cap];
r3 =
[3;4;r_in+l_a+l_e;r_in+l_a+l_e+l_c;r_in+l_a+l_e+l_c;r_in+l_a+l_e;0;0;h_cap;h_cap];

gd = [r1,r2,r3];
ns = char('Anode','Electrolyte','Cathode')';
sf = 'Anode + Electrolyte + Cathode';
[g,~] = decsg(gd,sf,ns);
geometryFromEdges(model,g);

figure; pdegplot(model,'EdgeLabels','on','FaceLabels','on'); axis equal;
title('Supercapacitor Geometry');
```




```
%% 4. COEFFICIENTS
specifyCoefficients(model,'m',0,'d',@assignD,'c',@assignC,'a',0,'f',0);
```

```
%% 5. BOUNDARY CONDITIONS
applyBoundaryCondition(model,'dirichlet','Edge',8,'u',0); % cathode grounded
J_anode = I_cell/(2*pi*r_in*h_cap);
applyBoundaryCondition(model,'neumann','Edge',1,'g',J_anode); % anode current input

%% 6. MESH
generateMesh(model,'Hmax',h_cap/20);

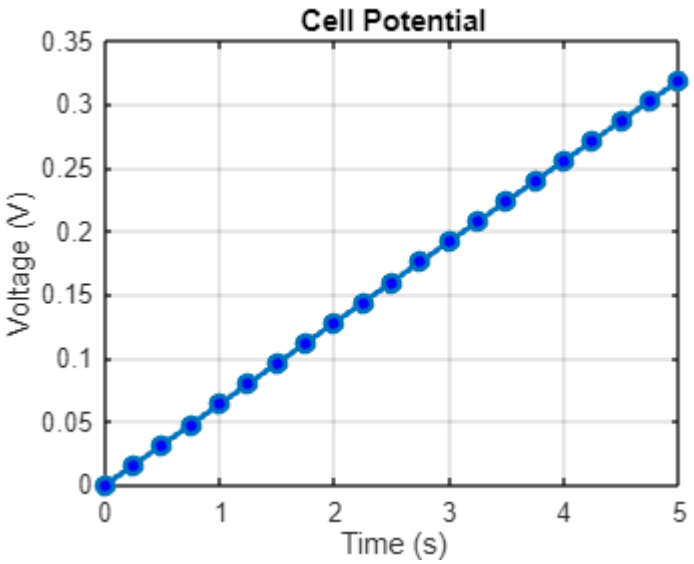
%% 7. INITIAL CONDITIONS
setInitialConditions(model,1e-6);

%% 8. SOLVE TRANSIENT
tlist = 0:0.25:5;
results = solvepde(model,tlist);
u = results.NodalSolution;

%% 9. POST-PROCESSING

% ---- Cell potential at anode (Face 1) ----
anode_nodes = findNodes(model.Mesh,'region','Face',1);
phi_anode = mean(u(anode_nodes,:),1);

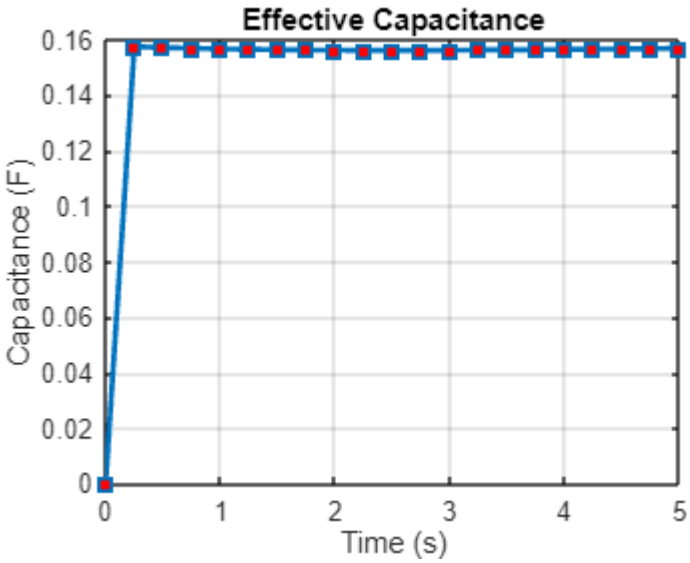
figure;
plot(tlist,phi_anode,'-o','LineWidth',2,'MarkerFaceColor','b');
grid on; xlabel('Time (s)'); ylabel('Voltage (V)'); title('Cell Potential');
```




```
% ---- Effective capacitance ----
Q = cumtrapz(tlist,I_cell*ones(size(tlist)));
C_eff = Q ./ phi_anode;

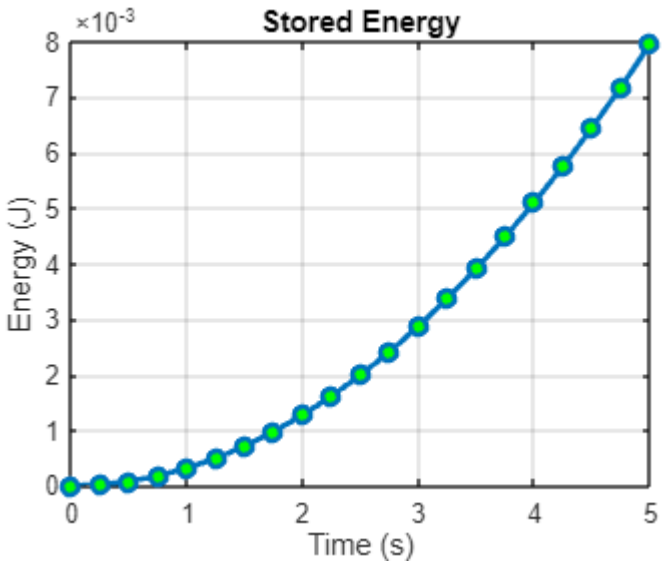
figure;
plot(tlist,C_eff,'-s','LineWidth',2,'MarkerFaceColor','r');
```

```
grid on; xlabel('Time (s)'); ylabel('Capacitance (F)'); title('Effective
Capacitance');
```




```
% ---- Stored energy ----
E_stored = 0.5 * C_eff .* phi_anode.^2;
figure;
plot(tlist,E_stored,'-o','LineWidth',2,'MarkerFaceColor','g');
grid on; xlabel('Time (s)'); ylabel('Energy (J)'); title('Stored Energy');

xlim([0.00 5.00])
ylim([0.0000 0.0080])
```




```
fprintf('Final capacitance: %.4f F\n',C_eff(end));
```

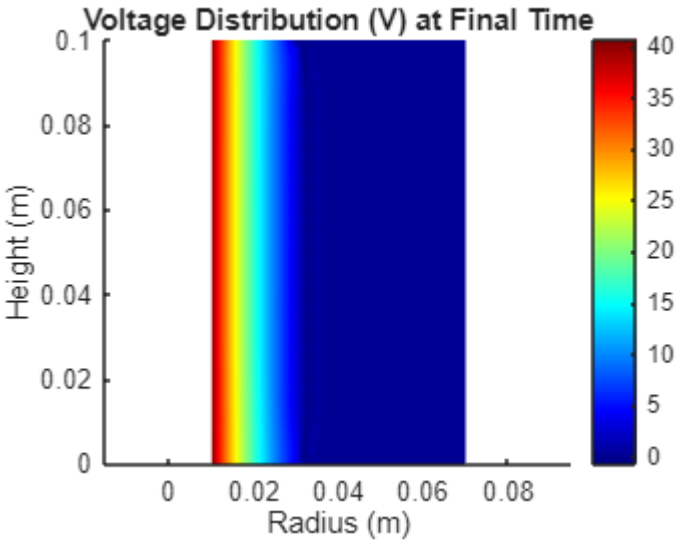
```
Final capacitance: 0.1569 F
```

```
fprintf('Final stored energy: %.4f J\n',E_stored(end));
```

```
Final stored energy: 0.0080 J
```

```
%% ---- 2D Voltage and Energy Density Maps (Final Time) ----
phi_final = u(:,end);

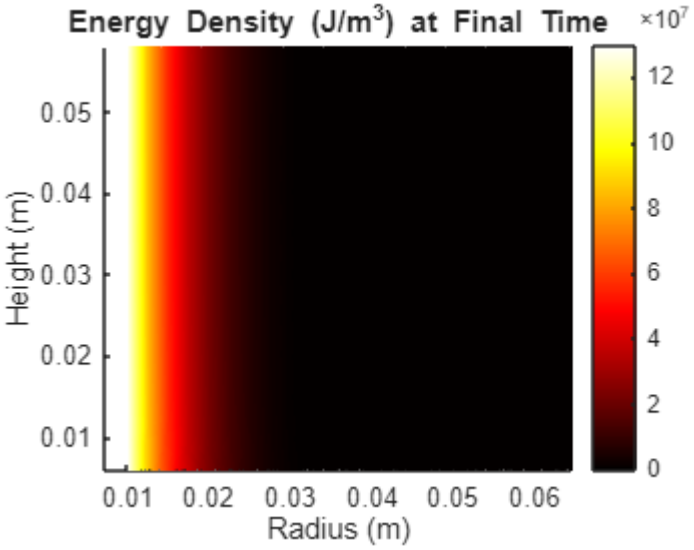
% Voltage distribution
figure;
pdeplot(model,'XYData',phi_final,'Mesh','off','ColorMap','jet');
axis equal; colorbar;
title('Voltage Distribution (V) at Final Time');
xlabel('Radius (m)'); ylabel('Height (m)');
```




```
% Energy density distribution
C_dl_map = zeros(size(phi_final));
C_dl_map(model.Mesh.Nodes(1,:) <= r_in+l_a) = C_dl*A_v;          % Anode
C_dl_map(model.Mesh.Nodes(1,:) > r_in+l_a+l_e) = C_dl*A_v;       % Cathode
C_dl_map((model.Mesh.Nodes(1,:) > r_in+l_a) & (model.Mesh.Nodes(1,:) <=
r_in+l_a+l_e)) = 1e-12; % Electrolyte

E_density = 0.5 * C_dl_map .* phi_final.^2;

figure;
pdeplot(model,'XYData',E_density,'Mesh','off','ColorMap','hot');
axis equal; colorbar;
title('Energy Density (J/m^3) at Final Time');
xlabel('Radius (m)'); ylabel('Height (m)');
```

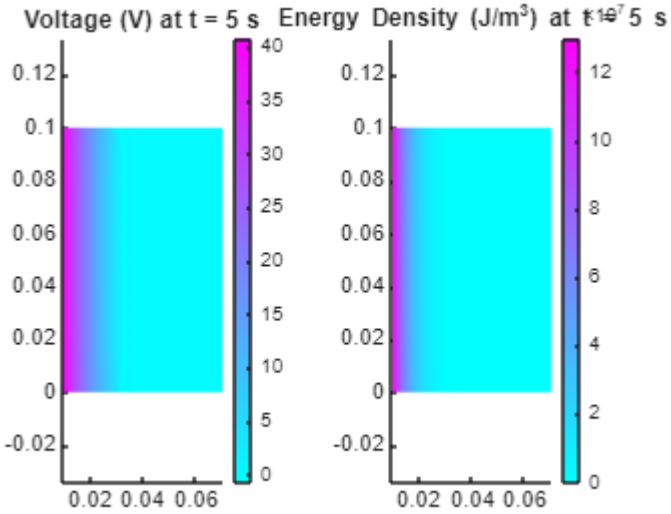



```
%% ---- Optional Animation Over Time ----
figure;
for k = 1:length(tlist)
    phi = u(:,k);
    E = 0.5 * C_dl_map .* phi.^2;

    subplot(1,2,1);
    pdeplot(model,'XYData',phi,'Mesh','off'); axis equal; colorbar;
    title(['Voltage (V) at t = ', num2str(tlist(k)),' s']);

    subplot(1,2,2);
    pdeplot(model,'XYData',E,'Mesh','off'); axis equal; colorbar;
    title(['Energy Density (J/m^3) at t = ', num2str(tlist(k)),' s']);

    drawnow;
end
```



```
%% ====== Helper functions ======
function d = assignD(region,~)
    r = region.x;
    n = numel(r);
    d = zeros(1,n);

    isAnode = region.subdomain==1;
    isElectrolyte = region.subdomain==2;
    isCathode = region.subdomain==3;

    d(isAnode | isCathode) = 0.13*1.2e6;  % C_dl*A_v
    d(isElectrolyte) = 1e-12;             % very small for solver

    d = d .* r;  % axisymmetric scaling
end

function c = assignC(region,~)
    r = region.x;
    n = numel(r);
    c = zeros(1,n);

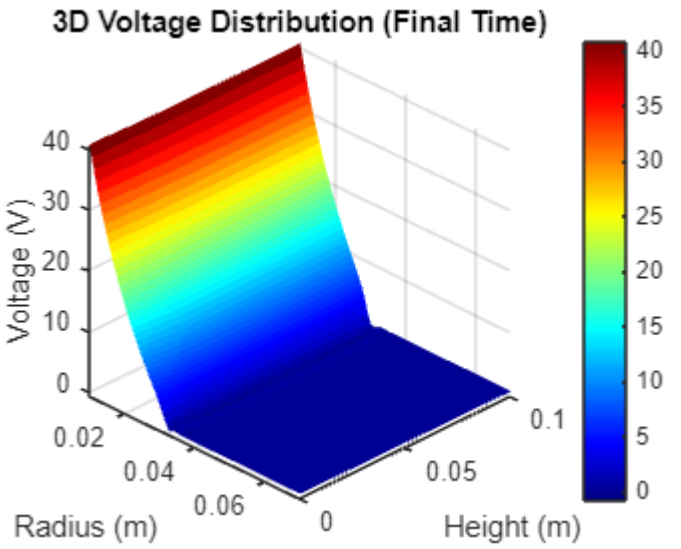
    sigma_s = 1e-3; sigma_l = 1e-1;
    eps_s = 0.75; eps_l = 0.6;

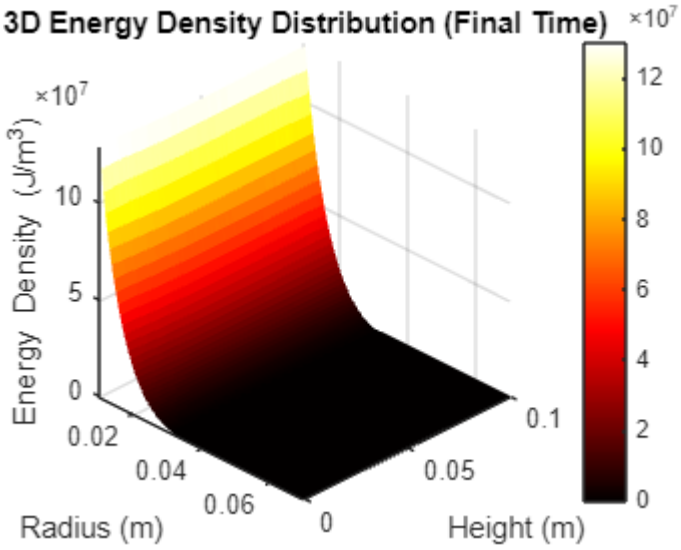
    isAnode = region.subdomain==1;
    isElectrolyte = region.subdomain==2;
    isCathode = region.subdomain==3;

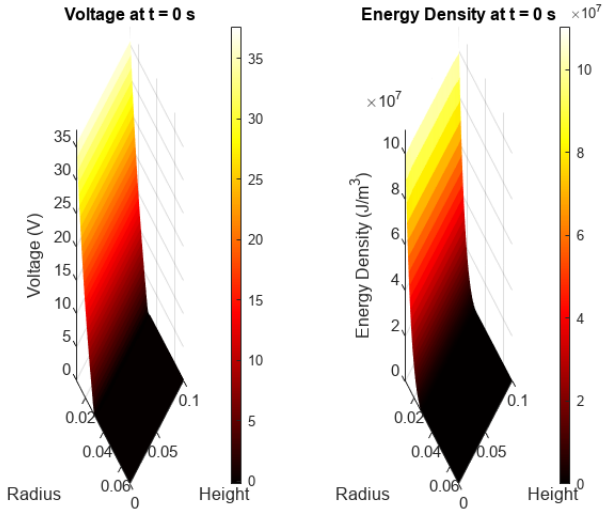
    c(isAnode | isCathode) = sigma_s * eps_s^1.5;
    c(isElectrolyte) = sigma_l * eps_l^1.5;

    c = c .* r;  % axisymmetric scaling
end

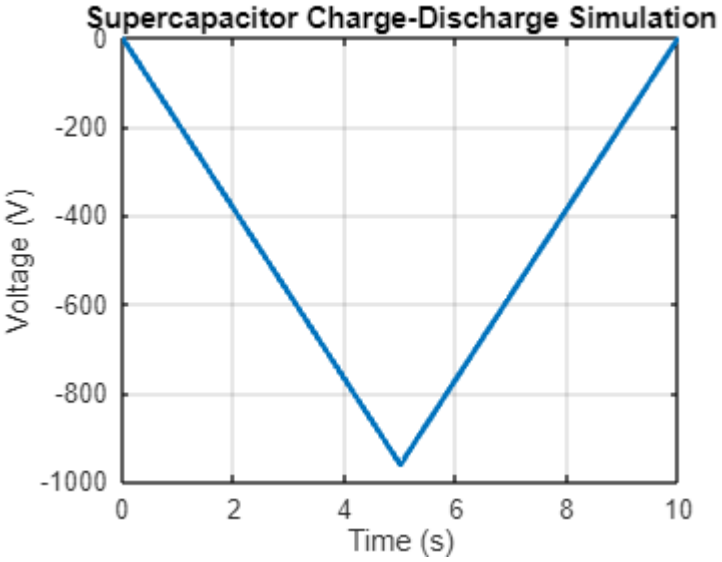
plot3DSupercapacitor(model,results.NodalSolution,tlist,C_dl,A_v,r_in,l_a,l_e);
```

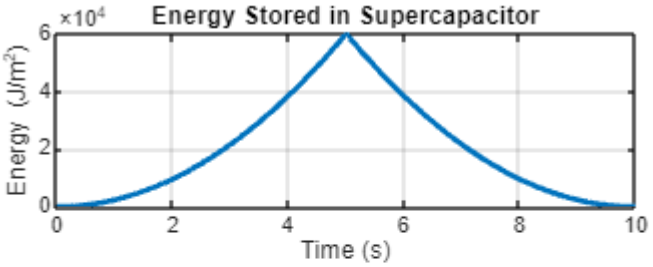


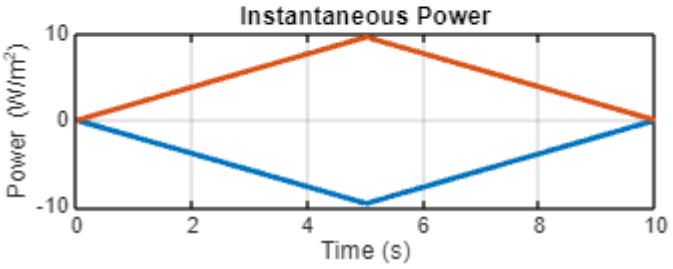


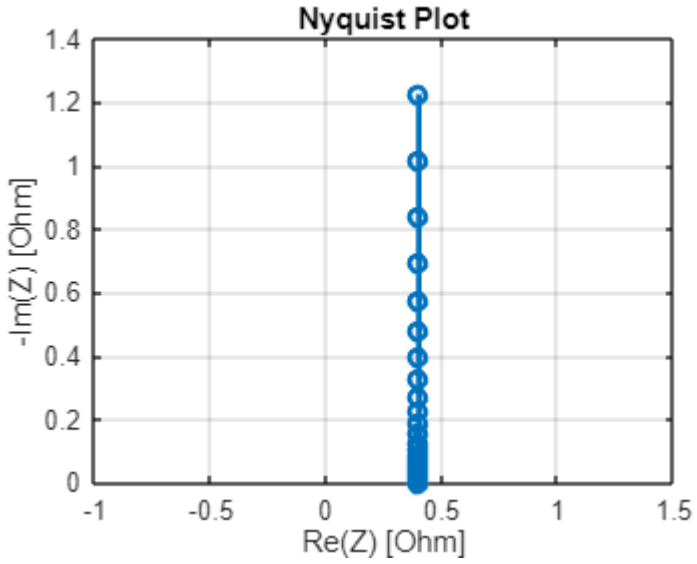


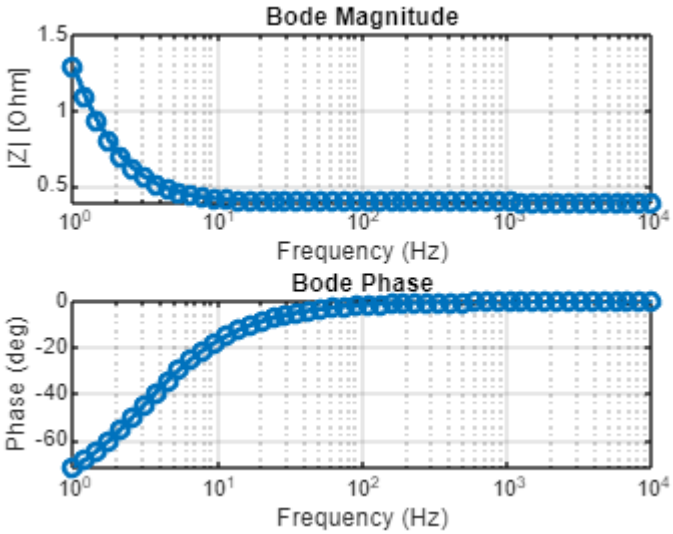

```
simulateSupercapacitorFull();
```


Supercapacitor Charge-Discharge Simulation
Voltage (V)
Time (s)
0
-200
-400
-600
-800
-1000
0
2
4
6
8
10


Energy Stored in Supercapacitor
×10^4
Energy (J/m^2)
Time (s)
6
4
2
0
0
2
4
6
8
10


Instantaneous Power
Power (W/m^2)
Time (s)
10
0
-10
0
2
4
6
8
10


Nyquist Plot
-Im(Z) [Ohm]
Re(Z) [Ohm]
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-1
-0.5
0
0.5
1
1.5




```
function plot3DSupercapacitor(model,u,tlist,C_dl,A_v,r_in,l_a,l_e)
% ========================================================================
% plot3DSupercapacitor - Plots 3D voltage and energy density for a
% supercapacitor PDE solution.
%
% Inputs:
%   model  - PDE model object
%   u      - Nodal solution (Nnodes x NtimeSteps)
%   tlist  - Vector of time steps
%   C_dl   - Double-layer capacitance (F/m^2)
%   A_v    - Electrode specific surface area (1/m)
%   r_in   - Inner radius of anode (m)
%   l_a    - Anode thickness (m)
%   l_e    - Electrolyte thickness (m)
%
% Usage:
%   plot3DSupercapacitor(model,u,tlist,C_dl,A_v,r_in,l_a,l_e)
% ========================================================================

nodes = model.Mesh.Nodes;
x = nodes(1,:)'; % radius
y = nodes(2,:)'; % height

% Determine C_dl map per node
C_dl_map_nodes = zeros(size(x));
C_dl_map_nodes(x <= r_in+l_a) = C_dl*A_v;                % Anode
C_dl_map_nodes(x > r_in+l_a+l_e) = C_dl*A_v;             % Cathode
C_dl_map_nodes((x > r_in+l_a) & (x <= r_in+l_a+l_e)) = 1e-12; % Electrolyte
```

```
% Create fine grid for smooth plotting
[rq,zq] = meshgrid(linspace(min(x),max(x),100), linspace(0,max(y),100));

%% 1. Voltage at final time
phi_final = u(:,end);
phi_grid = griddata(x,y,phi_final,rq,zq,'cubic');

figure;
surf(rq,zq,phi_grid,'EdgeColor','none');
colormap(jet); colorbar;
xlabel('Radius (m)'); ylabel('Height (m)'); zlabel('Voltage (V)');
title('3D Voltage Distribution (Final Time)');
view(45,30); axis tight; grid on;

%% 2. Energy density at final time
E_grid = griddata(x,y,0.5*C_dl_map_nodes.*phi_final.^2,rq,zq,'cubic');

figure;
surf(rq,zq,E_grid,'EdgeColor','none');
colormap(hot); colorbar;
xlabel('Radius (m)'); ylabel('Height (m)'); zlabel('Energy Density (J/m^3)');
title('3D Energy Density Distribution (Final Time)');
view(45,30); axis tight; grid on;

%% 3. Optional animation over time
animate = true; % Set to false to skip animation
if animate
    figure;
    for k = 1:length(tlist)
        phi = u(:,k);
        E = 0.5*C_dl_map_nodes.*phi.^2;
        phi_grid = griddata(x,y,phi,rq,zq,'cubic');

        subplot(1,2,1);
        surf(rq,zq,phi_grid,'EdgeColor','none'); colormap(jet); colorbar;
        xlabel('Radius'); ylabel('Height'); zlabel('Voltage (V)');
        title(['Voltage at t = ', num2str(tlist(k)),' s']);
        view(45,30); axis tight; grid on;

        subplot(1,2,2);
        E_grid = griddata(x,y,E,rq,zq,'cubic');
        surf(rq,zq,E_grid,'EdgeColor','none'); colormap(hot); colorbar;
        xlabel('Radius'); ylabel('Height'); zlabel('Energy Density (J/m^3)');
        title(['Energy Density at t = ', num2str(tlist(k)),' s']);
        view(45,30); axis tight; grid on;

        drawnow;
    end
end
```

```
end
```

```
function simulateSupercapacitorFull()
% ========================================================================
% FULL Supercapacitor Analysis:
% 1D PDE (transient voltage) + AC impedance + energy/power + charge-discharge
% ========================================================================

%% ------------------ PARAMETERS ------------------
L = 400e-6;      % Total thickness [m]
sigma = 1e-3;    % Effective conductivity [S/m]
Cdl = 0.13;      % Double-layer capacitance [F/m^2]
V0 = 0;          % Initial voltage [V]

% Charge-discharge settings
I_charge = 0.01;  % Charging current [A]
I_discharge = -0.01; % Discharging current [A]
t_cycle = 5;      % Duration of each charge/discharge [s]

%% ------------------ SPATIAL & TEMPORAL GRID ------------------
x = linspace(0,L,50);

% Charge-Discharge time vector
t_charge = linspace(0,t_cycle,50);
t_discharge = linspace(0,t_cycle,50);

%% ------------------ CHARGE CYCLE ------------------
sol_charge = pdepe(0, @(x,t,V,dVdx) pdefun(x,t,V,dVdx,sigma,Cdl),...
                   @(x) ic(x,V0),...
                   @(xl,Vl,xr,Vr,t) bc(xl,Vl,xr,Vr,t,I_charge),...
                   x,t_charge);
V_charge = sol_charge(:,:,1);

%% ------------------ DISCHARGE CYCLE ------------------
V_init_discharge = V_charge(end,:)'; % last voltage of charge
sol_discharge = pdepe(0, @(x,t,V,dVdx) pdefun(x,t,V,dVdx,sigma,Cdl),...
                      @(x) ic(x,V_init_discharge),...
                      @(xl,Vl,xr,Vr,t) bc(xl,Vl,xr,Vr,t,I_discharge),...
                      x,t_discharge);
V_discharge = sol_discharge(:,:,1);

%% ------------------ CONCATENATE RESULTS ------------------
V_total = [V_charge; V_discharge];
t_total = [t_charge t_cycle + t_discharge];
V_avg = mean(V_total,2); % average across thickness

%% ------------------ PLOT VOLTAGE ------------------
figure;
plot(t_total,V_avg,'LineWidth',2);
```

```
xlabel('Time (s)'); ylabel('Voltage (V)');
title('Supercapacitor Charge-Discharge Simulation');
grid on;

%% ------------------ ENERGY & POWER ------------------
Energy = 0.5 * Cdl * V_avg.^2;      % Energy stored (J/m^2)
Power = V_avg .* [I_charge*ones(size(t_charge))
I_discharge*ones(size(t_discharge))];

figure;
subplot(2,1,1);
plot(t_total,Energy,'LineWidth',2);
xlabel('Time (s)'); ylabel('Energy (J/m^2)');
title('Energy Stored in Supercapacitor');
grid on;

subplot(2,1,2);
plot(t_total,Power,'LineWidth',2);
xlabel('Time (s)'); ylabel('Power (W/m^2)');
title('Instantaneous Power');
grid on;

%% ------------------ AC IMPEDANCE ------------------
freqs = logspace(0,4,50); % 1 Hz to 10 kHz
R = L/sigma;              % ESR approx
C = Cdl;                  % Capacitance approximation
Z = R + 1./(1i*2*pi*freqs*C);

% Nyquist
figure;
plot(real(Z),-imag(Z),'-o','LineWidth',2);
xlabel('Re(Z) [Ohm]'); ylabel('-Im(Z) [Ohm]');
title('Nyquist Plot');
grid on;

% Bode
figure;
subplot(2,1,1);
semilogx(freqs,abs(Z),'-o','LineWidth',2);
xlabel('Frequency (Hz)'); ylabel('|Z| [Ohm]');
title('Bode Magnitude');
grid on;

subplot(2,1,2);
semilogx(freqs,angle(Z)*180/pi,'-o','LineWidth',2);
xlabel('Frequency (Hz)'); ylabel('Phase (deg)');
title('Bode Phase');
grid on;

end
```

```
%% ====== PDE FUNCTIONS ======
function [c,f,s] = pdefun(~,~,V,dVdx,sigma,Cdl)
% V is a column vector of length Nx
Nx = numel(V);
c = Cdl*ones(Nx,1);
f = sigma*dVdx; % column vector of same size as V
s = zeros(Nx,1);
end

function V0 = ic(~,Vinit)
% Initial condition: column vector
if numel(Vinit) > 1
    V0 = Vinit(:); % ensure column
else
    V0 = Vinit;
end
end

function [pl,ql,pr,qr] = bc(~,Vl,~,Vr,~,I_app)
% Boundary conditions: Anode current input, cathode grounded
pl = -I_app*ones(size(Vl)); ql = ones(size(Vl));
pr = zeros(size(Vr)); qr = ones(size(Vr));
end
```